

Workshop on Moralizing Radio Frequency Holography for Palliative Care via Value-Sensitive Design

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Abstract

Palliative care provides healthcare treatment to patients with critical and terminal medical conditions at the end of their lives. This type of care is usually provided in highly medicalised settings, given the current health provision at hospitals, depriving many patients of the possibility to spend their final moments at home with their loved ones. At home, radio frequency (RF) holography is considered a possible solution or assistant for these concerns. Radio frequency holography utilises electromagnetic waves, typically used for Wi-Fi, to gain a deeper understanding of occurrences within an environment by applying computational reasoning, such as machine learning, to visualise or label feedback. This new technology has the potential to disrupt social norms and standards and change the phenomenology of palliative care for terminal residents, healthcare professionals, and family members alike. This paper presents a methodology to apply value-sensitive design (VSD) to RF holography for palliative care to help engineers gain tools and the vocabulary to better communicate, understand, and address ethical concerns in the technologies they build.

Ccs concepts

Computing/technology policy, Life and medical sciences, Human and societal aspects of security and privacy, Law, social and behavioral sciences, Ubiquitous and mobile computing

Keywords

RF Holography, Ethics, Value Sensitive Design, Palliative Care

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1 Introduction

This paper overviews the case study of using radio frequency electromagnetic waves, used for telecommunications, for palliative care. It pays close attention to how to create ethical solutions in this domain of newly emerging and potentially socially disruptive technology to help near-end-of-life patients. This introduction covers important background information on phenomenology in the health sector, general ethical considerations and presents a general overview on the technology itself. Section two overviews why doing ethics in this domain is so challenging and offers Value Sensitive Design (VSD) as a potential strategy for ethical development. The third section provides a more detailed overview of the case study at Královské Vinohrady Hospital in Prague that will be run in 2025-2026. Section four outlines a model for generating values and design requirements from stakeholders in a conference setting to assist engineers in considering the phenomenology of palliative care residents and decision-making based on moral values. In conclusion, it is found that VSD alone is not enough to ensure ethical development, but it is a beneficial starting point, especially in its ability to align stakeholders around values.

1.1 Phenomenological Bioethics

Phenomenological bioethics, which stems from 20th-century phenomenology, centers on exploring human experiences such as pain and empathy, especially in healthcare contexts. It suggests that truly understanding patients' lived experiences gives deeper

insight into health-related challenges than simply relying on broad ethical rules. This approach also emphasizes how people relate to their own health conditions and recognizes the influence of societal views. Furthermore, it encourages healthcare technology creators to genuinely consider what patients experience and need.

1.2 Phenomenology in Palliative Care

Phenomenology is recognized as an effective method for advancing palliative care [4,5]. This approach is often implemented through qualitative interviews, referred to as phenomenological interviews, which may involve patients or healthcare professionals. Such interviews are typically open-ended and lack a predetermined structure or content, yet consistently aim to elucidate the lived experiences of both patients and healthcare providers within medical and nursing contexts.

Within palliative care research, these methods have been utilized to gain a deeper understanding of patient comfort by analyzing recurring themes in participants' responses to targeted questions regarding their healthcare experiences. For example, Coelho et al. [5] found that fostering humanized care and addressing existential concerns such as hope and interpersonal relationships had a significant positive impact on patient comfort. The study also identified additional contributors to comfort, including the creation of differentiated care environments and effective symptom management.

Although phenomenology does not constitute a medical discipline, it can be integrated into healthcare processes to provide valuable insights. One practical application involves phenomenological interviews [6,5], whose critical perspective is especially beneficial for identifying social biases or stigma associated with specific medical conditions or treatment pathways. These interviews seek to uncover both individual patient experiences and broader societal attitudes concerning health and illness. In the context of palliative care, such interviews may facilitate the establishment of realistic goals for end-of-life care, inform the adaptation of home care strategies, and support patients in navigating challenging circumstances.

1.3. Ethical Challenges in Artificial Intelligence for Healthcare

Research demonstrates that artificial intelligence (AI) offers considerable promise in the field of palliative care, with mortality prediction emerging as one of its most prevalent applications [7]. Nonetheless, palliative care is fundamentally holistic, necessitating attention to patients' needs that extend beyond mortality forecasting. The delivery of effective care requires adherence to patients' preferences and a comprehensive approach to their overall well-being. Furthermore, the use of machine learning (ML) in electronic health records (EHR) constitutes another significant AI application that supports optimal patient care pathways. ML algorithms can monitor EHRs to identify individuals who may have less than 12 months to live, thereby facilitating timely initiation of advanced care planning [8]. In addition, AI-enabled remote health monitoring allows for regular

health assessments of palliative care patients in home settings, enhancing access to clinical data and enabling the development of personalised treatment strategies.

The examples mentioned above underscore the importance of designing AI systems that promote patient autonomy:

"[...] phenomenological approaches may help us to develop hypotheses regarding the relationship between particular styles of palliative care provision and the unique experiences of very sick people and those close to them" [4, p. 131].

Ethical challenges associated with AI in palliative care are numerous. Concerns about privacy and informed consent are prominent, as AI applications in healthcare require access to extensive sensitive patient data:

"[...] there is an additional risk that such companies gain undue influence and impose their own values and incentives (eg, profit maximisation) that are not congruent with, and even run counter to, long-standing values in healthcare (eg, trust)" [9, p. 4].

Another significant ethical concern is the possible erosion of empathy and personal interaction as automation becomes more prevalent [8]. Phenomenology addresses this issue by focusing on intersubjectivity—especially empathy and listening to patient experiences—in healthcare relationships. This discussion prompts consideration of whether artificial intelligence should serve as a full replacement for, or merely a supplement to, human care in contexts involving vulnerable individuals, such as those receiving palliative care. Given the importance that technology users place on meaningful human contact [9,5], comprehensive replacement is generally viewed unfavorably. The integration of AI into healthcare must be designed to avoid replicating societal challenges or further disadvantaging vulnerable populations [10]. Kathleen Lynch's cautionary perspective remains highly relevant [11]:

"[...] assistive technologies can complement personal care; however, they cannot replace it completely. Helping someone with washing or showering, with dressing, getting up, going to bed or going to the toilet is not just a matter of physical assistance, though it is that. It also involves respecting them, as well as encouraging, reassuring, supporting and explaining; how important these care dispositions are in any given day or event will vary with the person involved." (pp. 89-90)

1.4 HOLDEN Technology

The European Innovation Council (EIC) grant, "Ethical Design of Holography with Dense Wireless Networks" (<https://holden-project.eu/>), otherwise known as HOLDEN, has the goal of ethically developing 3D vision utilizing dense ambient radio frequency electromagnetic waves typically used for WIFI. This technology can be adapted for a variety of use cases, including person/object detection, activity recognition, gesture recognition, industrial robotics, smart TVs and elderly and palliative care. Each

of these contexts has been explored within the HOLDEN consortium.

RF holography is an evolving field that has several unique features: RF waves for Wi-Fi and 5G are continuously more densely covering inhabited areas, making it ubiquitous; it can repurpose pre-existing telecommunication infrastructure, it works in the dark and through walls, and can gather sensitive data (such as sleep data and breath and heartrate) without the need for wearables. Each of these features has the potential to create useful applications but also runs the risk of infringing on basic rights such as privacy and autonomy if not carefully implemented with proper value alignment and corresponding technical requirements.

2 Ethics of New and Emerging Technology

2.1 Socially Disruptive Technology

One challenge we face is determining how to practically embed ethics in RF technology for palliative care while the technology is still in development. RF Holography is by definition an “emerging technology,” characterized as new, in development, and likely to have a significant socioeconomic impact [12]. RF Holography, even in this specific context, has the potential to shift moral norms. When a technology becomes embedded in the world, it changes the expectations for what ought to occur. This is because “ought implies can” – in order to have the expectation to do something, it must first be possible. For example, the primary reason one should call for an ambulance in an emergency situation is because it is feasible; phones, cars, dispatch systems and ambulances now exist, so there is a moral duty to utilize them.

We must preemptively design policy and technology to promote ethical use in an uncertain future. It has been established that designs are not neutral but instead are embedded with values that reflect the priorities (or lack thereof) of their designers [13]. Unfortunately, designers and engineers are not always aware of the needs of their users and might prioritize economic values over moral ones. Furthermore, it is not uncommon for baseline moral principles upon commercialization to centralize not breaking the law over having a positive impact. While this is generally abhorrent, it is inexcusable in the palliative care context.

2.2. Value Sensitive Design (VSD)

Bataya Friedman and colleagues first introduced VSD in the late 1980s as a method in the field of Human-Computer Interaction [14]. It was becoming evident that software systems were forming a foundation of critical infrastructure, and their implementation would therefore impact not just people but also future systems. Whether intentionally or not, the values embedded into these systems (such as efficiency over accuracy) would have iterative impacts as the tools spread and were built upon.

VSD is central to our approach. While there are now a variety of methods, in short, VSD seeks to first acknowledge potential

limitations of a technology and possible (im)moral normative effects and determine essential values that must be upheld to build design constraints around what can be modulated [15]. Determining essential values for emerging technologies can be challenging as there are a variety of approaches for determining ethical concerns including “generic, anticipatory, risk, experimental and participatory and deliberative approaches” [12]. Some values might juxtapose with each other. To build ethical technology, moral values should be prioritized. Value-sensitive design can and should be an iterative process as the technology develops and potential ethical dilemmas develop.

3 RF-Sensing in Palliative Care Context

3.1 Case Study

To support the design of value-sensitive technologies, user experience is considered important for identifying general concerns that developers may not address during the design process. In palliative care research, studies have investigated how patient comfort can be understood through interviews with patients by analyzing recurring themes in their responses to questions about their healthcare experiences. For example, Coelho and colleagues [5] found that focusing on humanising care and addressing existential topics such as hope and relationships could improve patient comfort. Including these experiences in Value Sensitive Design (VSD) can help tailor medical treatment and care to better align with patient needs, promoting autonomy and informed decision-making.

RF-sensing technology enables continuous, non-intrusive monitoring of individuals, such as palliative care patients in their homes, without requiring physical contact or wearable devices. This technology can support ongoing health monitoring, including heart rate analysis, activity tracking, and mobility detection, which may inform various medical interventions as needed.

RF-sensing acts as a mediating technology, providing additional insights into patients and their medical conditions. It changes the way patients interact with their home environment, allowing it to serve as a place for less invasive treatment. This reassignment of the home as a site for care can facilitate patient comfort within the context of their condition, compared to traditional hospital settings, and potentially allows patients greater control over their final days. Being at home also enables more frequent visits from loved ones and preserves a higher degree of autonomy than being hospitalized.

Recognizing potential future applications of RF-sensing in alternative medical services, an ethical assessment of its current impact is necessary. VSD of RF-sensing should consider patients' experiences and values regarding receiving healthcare at home, including aspects important to them and ways to improve their mental and physical well-being. Incorporating user values and experiences into the technology is an ongoing process; therefore, the following workshop aims to explore various design approaches to appropriately integrate values into the development of RF-sensing for HOLDEN.

3.2 Ethical Challenges

We have identified the following key ethical values that need to be applied for the design of the specific prototype that will be used for the testing period of the RF-sensing technology in palliative care.

- The technology should be non-invasive and private.
- The technology should be reliable to the extent that the data obtained has minimal margins for error.
- The safety of the product is key. In no scenario should the product be harmful to patients, nor should it hinder personal values such as freedom and autonomy.
- Freedom and autonomy of patients should be promoted by the technology. The technology should also aim to enhance the patient experience of quality of life, autonomy and freedom at home (as opposed to being constrained in a hospital situation)

Among the ethical benefits of in-home palliative care are the improved quality of life for patients. This can take several different shapes. First of all, patients feel more comfortable at home at the end of life, where they have better access to emotional and spiritual support provided by relatives or close ones. Secondly, generally speaking, people prefer to die at home and not in the hospital, which speaks in favour of home-based palliative care. Finally, not only can the quality of life of patients improve in in-home care scenarios. Also, the patient's relatives and close ones could prefer their loved ones to be close to fully support them at the end of life.

4 Example Stakeholder Workshop for Case Study

In this section, we briefly present an example VSD workshop and demonstrate how it may be used to align designers, engineers, and executives to more ethically build RF Palliative Care solutions. A version of this workshop was carried out at the annual HOLDEN Plenary meeting in Padova, Italy, June 2025.

4.1 Methodology

Type: Open workshop

Length: 60-80 minutes

Participants: 6-30 people

The workshop begins with a brief overview of essential concepts for VSD, namely an understanding of what values are and the distinction between moral and non-moral values. Additionally, the workshop illustrates how values are embedded into all technologies, even when not taken explicitly into consideration.

Next, the palliative care case study is introduced, including phenomenological findings. Participants consider the most essential values for this context, centering on the relationship between values and user experience (including other stakeholders such as loved ones and the healthcare team).

Following, groups of 4-5 members come up with design requirements and/or regulations to ensure that the chosen value standards are met. We will consider whether these choices still

uphold the prioritized values when the context shifts and when all the values and requirements are combined into a singular system.

4.2 Expected results

By the end of the workshop, participants will come away with a better understanding of the challenges and opportunities in palliative care as well as the potential applications of RF holography. More generally, the workshop has three expected outcomes: first, we aim to raise awareness among participants regarding the values embedded in technologies during the development process; second, we hope participants feel more empowered to address future ethical concerns when developing other tools; third, we hope to gain a better understanding of alternative design paths for the RF-sensing technology for palliative care currently under development in HOLDEN.

5 Conclusion

RF-sensing, when done well, can offer patients the opportunity to receive quality care in their own homes during their final moments. However, it requires thoughtful consideration to increase the likelihood of the best outcome. Value-sensitive design is a tool that prioritizes moral values, sets requirements based on those values and uses those requirements to create technologies where those standards are met. It is effective at making clear the priorities of a technology for all stakeholders and building towards those goals. Working proactively to consider what values are most important and how those translate to requirements can be a challenging task that necessitates a phenomenological approach to, in the case of RF sensing in palliative care, understand the needs of patients, their loved ones, and also their care providers. Furthermore, building technical requirements and designs requires the expertise and moral awareness of technologists who understand how certain design decisions will affect the capabilities. New and emerging technologies have the added challenge of potential spin-off applications further shifting societal norms and values in ways that might not be immediately predictable. Therefore, even before applying value sensitive design, it may be helpful to also consider other philosophical approaches that are currently being embraced by HOLDEN such as scenario building [16], mediation theories [17,18,19], and guidance ethics [20] to help ensure that relevant issues are being addressed by VSD.

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